

DYNAMIC FINITE ELEMENT ANALYSIS OF STRUCTURES

Mechanical Engineering 2024-2025 Master semester 2

Specialization/orientation: Mechanics of solids and structures

CFU: 3

Language: English

Semester: Spring

Lectures: 2 hours per week x 14 weeks

Exercises: 1 hour per week x 14 weeks

Exam: summer session (written)

Summary. The course covers linear elastic numerical analysis of structures using the finite element method. Students will learn to master numerical methods commonly used in static and dynamic structural analysis, applying these techniques to solve realistic problems in both academic and industrial contexts.

Content.

- Strong and weak forms of the governing equations in solid mechanics
- Extension of the finite element method to 3D dynamic structural problems
- Development of solid and shell finite elements
- Strain matrix, local and global coordinate systems, assembly of local elements
- Development of modal parameter extraction methods
- Numerical analysis of the forced regime of structures
- Examples and case studies in MATLAB

Learning Prerequisites.

- Finite element method
- Finite element modelling and simulation
- Structural mechanics
- Solid mechanics
- Mechanical vibrations

Learning outcomes. By the end of the course, the student must be able to:

- Formulate and solve a problem in numerical structural analysis.
- Develop a critical awareness of the sources of errors inherent to the FE modelling practice.
- Use an appropriate multi-physics modelling and simulation to solve a complex physical problem.

Transversal skills.

- Use a work methodology appropriate to the task.
- Plan and carry out activities in a way which makes optimal use of available time and other resources.
- Use both general and domain specific IT resources and tools.
- Write a scientific or technical report.

Teaching methods. Ex cathedra lectures, exercises in the classroom and computer lab sessions.

Expected student activities. Solution of assignments, active participation in case studies (in-class exercises)

Assessment methods. 40% mini-projects during the semester and 60% final written exam

Ressources.

- Thomas Gmür, Méthode des éléments finis en mécanique des structures, Presses polytechniques et universitaires romandes (PPUR), Lausanne, 2018, ISBN 978-2-88915-158-5.
- Thomas Gmür, Dynamique des structures – Analyse modale numérique, Presses polytechniques et universitaires romandes (PPUR), Lausanne, 2014 (2ème édition), ISBN 978-2-88074-813-5
- Thomas J. R. Hughes, The Finite Element Method – Linear Static and Dynamic Finite Element Analysis, Prentice-Hall, Englewood Cliffs, NJ, 1987 (Dover Publications, 2000), ISBN 978-0-48641-181-1